3 Electrostatics–III

CONDUCTORS

- A conductor has a lot of charges that are free to move.
- Consider an isolated conductor as shown in figure 3.1 (a).
- Due to the applied external field $\vec{E}_e$, positive free charges are all being pushed in the same direction as of the electric field. The negative free charges, on the other hand, goes in the opposite direction.
- This movement of charges takes place very rapidly. The free charges generally accumulates on the surface of the conductor and form an induced surface charge.
- The induced charges then creates an internal induced field $\vec{E}_i$ that cancels the field applied outside $\vec{E}_e$.
- The result is shown in the figure 3.1(b). Therefore, a perfect conductor cannot contain an electrostatic field within it.
- A conductor is referred to as an equipotential body because its potential is constant throughout the conductor.
- This is because $\vec{E} = -\nabla V = 0$. Another way of understanding this by Ohm's law $(\vec{J} = \sigma\vec{E})$.
- In a perfect conductor, to maintain a finite current density $(\vec{J})$ it will require the electric field to vanish inside the conductor.
  In other words, $\vec{E} \to 0$ in a perfect conductor.
- If some charges are added to a conductor, the charges will rise to the surface. They soon will organise themselves such that the field inside the conductor vanishes.
- According to Gauss's law, if $\vec{E} = 0$, the charge density (ρ_v) must be zero.
- It is concluded there is no electric field inside the conductor. Under static conditions, $\vec{E} = 0$,

Definitions

An **electrical conductor** is a substance in which electrical charge carriers, usually electrons, flow smoothly from atom to atom under the impact of applied voltage. Electrical conductivity is the reciprocal of electrical resistivity and outlines a material's ability to conduct the electric current.

Gray Matter Alert!!!

Pure silver is the best conductor of electricity, but it is rarely used over copper. Liquids like mercury, saturated salt-water the solution acts as a feasible electrical conductor.

$\rho_v = 0$ and $V_{ab} = 0$ inside a conductor where V_{ab} is the potential difference between points a and b in the conductor.

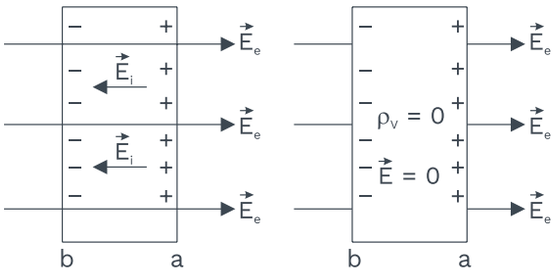


Figure 3.1: (a) Isolated conductor under the influence of an applied field. (b) The conductor has zero electric field under static conditions.

- Under the influence of the applied electric field, the available free electrons start to move.
- During their motion, electrons collide or strike with the atoms in their path and rebound in random directions. This process is regarded as the drifting of the electrons.
- The electron attains a constant average velocity referred to as the drift velocity (v_d) after some time of movement.
- **Drift current:** The current produced due to the drifting of such free electrons in metallic conductors.
- The drift velocity (v_d) is directly proportional to the applied electric field ($\vec{E}$).

  Mathematically, $v_d \propto \vec{E}$ .... (3.1)
- Resolving the proportionality sign,

  $v_d = -\mu_e \vec{E}$ ... (3.2)

  where μ = proportionality constant
  or referred to as the mobility of electrons in a given material.
- Mobility (μ_e) for electrons is generally positive.
- The negative sign in equation (3.2) denotes that the velocity of the electrons is in the opposite direction to the electric field ($\vec{E}$).

Definitions

In electromagnetism, the **current density** is the measure of charge per unit of time that flows by the unit cross-section area.

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The current density vector has the magnitude of the electric current flowing per cross-sectional area and the direction of the movement of the positive charges at a given point in space.

Thus, Mobility(μ) $= \frac{\text{Velocity}}{\text{Field}} = \frac{\text{m/ s}}{\text{V / m}} = \frac{\text{m}^2}{\text{Vs}}$

The S.I. unit of mobility is square meters per Volt-second (m^2/Vs).

Mobility for Aluminium $\Rightarrow \mu_{Al} = 0.0012$

Mobility for Copper $\Rightarrow \mu_{Cu} = 0.0032$

- The relation between current density (J) and velocity (v) can be written as-

$$J = \rho_v v \quad \ldots\ldots (3.3)$$

If the number of protons and electrons are equal and then the material is always electrically neutral.

So, $\rho_v = 0$ for the neutral materials.

Equation (3.3) can be re-written as-

$$J = \rho_e v_d \quad \ldots\ldots (3.4)$$

where ρ_e = charge density due to free electrons.

- The free electrons charge density (ρ_e) is evaluated by the product of charge 'q' on one electron and the number of free electrons per cubic meter.

Therefore, $\rho_e = nq$

where n = number of free electrons per cubic meter.

Substituting v_d in equation (3.4), we get -

$$\vec{J} = \rho_e \mu_e \vec{E} \quad \ldots (3.5)$$

Gray Matter Alert!!!

Free electron theory demonstrates that metals contain free electrons that are responsible for electrical conductivity and serve the laws of classical mechanics.

POINT FORM OF OHM'S LAW

- The relation between current density ($\vec{J}$) and electric field ($\vec{E}$) can be expressed as -

$$\vec{J} = \sigma\vec{E} \quad \ldots (3.6)$$

where σ = conductivity of the material.

- Equation (3.6) is for metallic conductors and conductivity (σ) is usually measured in siemens per meter (S/m).

Also, equation (3.6) is regarded as point form of Ohm's law.

Conductivity of Al, $\sigma_{Al} = 3.82 \times 10^7$ S/m

Conductivity of Cu, $\sigma_{Cu} = 5.8 \times 10^7$ S/m

- Conductivity is constant for a wide range of current densities and electric field intensities for the metallic conductors.
- In all the directions, metallic conductors have the same properties. Thus, it is isotropic in nature, and these materials obey the Ohm's law.
- On comparing equations (3.5) and (3.6), we get-

  $\sigma = \rho_e \mu_e$ ...(3.7)

  Equation (3.7) defines conductivity in terms of mobility and charge density of the free electrons.
- Resistivity (ρ_R) is defined as the reciprocal of the conductivity.

  $\rho_R = \frac{1}{\sigma}$ ... (3.8)

  Also, the conductivity (σ) is inversely dependent on the temperature (T).
- As the temperature increases, the vibration of atoms in the crystalline structure increases. With these increased vibrations of electrons, the drift velocity decreases, remove hence the mobility (μ) decreases which decreases the conductivity.
- In other words, if the temperature increases, the conductivity decreases and resistivity increases.
- Now, considering that the charge distribution is suddenly unbalanced inside the conductor. Then, the numerous electrons will try to remain inside the conductor.
- All the electrons are negatively charged and repel each other due to their own electric fields. These electrons get accelerated from each other and cause imbalance and reach at the surface of the conductor.
- Also, the conductor is surrounded by the insulating medium and the electrons that are driven from inside the conductor, remain at the surface of the conductor.

Gray Matter Alert!!!

A perfect dielectric is a material with zero electrical conductivity and exhibits only the displacement current. This stores and returns electrical energy like an ideal capacitor.

Therefore, the charge density is always zero within a conductor.

- At any location inside the conducting material, no charge or electric field can exist under static conditions. The charge can exist on the conductor's surface, and the charge distribution on the surface is determined by the conductor's shape.
- The conductivity of the ideal conductor is infinite. The conducting surface is an equipotential surface.

CONTINUITY EQUATION

- The key principle of conservation of charge says that the charges can neither be created nor destroyed, although an equal number of positive charges and negative charges can be produced by separation, or lost by recombination.

  ∴ The time rate of the decrease in charge within a given volume should be equal to the net current flow outwards through the surface of the volume.

- The outward current (I_{out}) coming out from the closed surface is the charge created simultaneously and expressed as-

$$I_{out} = \oint \vec{J} \cdot d\vec{S} = -\frac{dQ_{in}}{dt} \quad \ldots (3.9)$$

where, Q_{in} = Total charge enclosed by the closed surface.

Using divergence theorem, equation (3.9) can be re-written as-

$$\oint_s \vec{J} \cdot d\vec{S} = \int_v \nabla \cdot \vec{J} \, dV \quad \ldots(3.10a)$$

$$\text{But, } \frac{-dQ_{in}}{dt} = \frac{-d}{dt}\int_v \rho_v \, dV = -\int \frac{\partial \rho_v}{\partial t} dV \quad \ldots (3.10b)$$

Substituting equations (3.10a) and (3.10b) in equation (3.9), we get-

Previous Year's Questions

The continuity equation is a union of which of the following two laws?

(A) Gauss law and Ohm's law.
(B) Ohm's law and Ampere law.
(C) Gauss law and Ampere law.
(D) Maxwell law and Ampere law.

$$\int_v \nabla \cdot \vec{J}\, dV = -\int_v \frac{\partial \rho_v}{\partial t} dV$$

or $$\nabla \cdot \vec{J} = -\frac{\partial \rho_v}{\partial t} \quad \text{... (3.11)}$$

Equation (3.11) is referred to as continuity equation.

- This continuity equation is derived from the law of conservation of charge, which states that there will be no charge accumulation at any point.

Example 1:

The current passing through the hemispherical shell has a radius of 38 cm and a current density of $\vec{J} = \frac{1}{r^3}\left(2\cos\theta \hat{a}_r + \sin^2\theta \hat{a}_\theta\right) A/m^2$ where $0 < \theta < \frac{\pi}{2}$ and $0 < \phi < 2\pi$ is.

(A) 16.5 A (B) 15.5 A
(C) 20 A (D) 12.5 A

Solution:

Given,

The current density-

$$\vec{J} = \frac{1}{r^3}\left(2\cos\theta \hat{a}_r + \sin^2\theta \hat{a}_\theta\right) A/m^2$$

where $0 < \theta < \frac{\pi}{2}$ and $0 < \phi < 2\pi$.

Radius of the hemisphere r = 38 cm = 0.38 m

The current will be –

$$I = \int \vec{J} \cdot d\vec{S}$$

where $d\vec{S} = r^2 \sin\theta\, d\theta\, d\phi\, \hat{a}_r$

$$I = \int_{\theta=0}^{\frac{\pi}{2}} \int_{\phi=0}^{2\pi} \frac{1}{r^3}\left(2\cos\theta\, r^2 \sin\theta\, d\phi\, d\theta\right)$$

$$I = \int_{\theta=0}^{\frac{\pi}{2}} \int_{\phi=0}^{2\pi} \frac{2}{r}\cos\theta \sin\theta\, d\phi\, d\theta$$

Rack your Brain

How is quantum free electron theory dissimilar from classical free electron theory?

$$I = \frac{2}{r}(2\pi - 0)\int_{\theta=0}^{\frac{\pi}{2}} \cos\theta \sin\theta \, d\theta$$

$$I = \frac{4\pi}{r}\left[-\frac{1}{4}\cos 2\theta\right]_0^{\frac{\pi}{2}}$$

$$I = \frac{4\pi}{r}\left(-\frac{1}{4}\right)[\cos 2\theta]_0^{\frac{\pi}{2}}$$

$$I = -\frac{\pi}{r}(-1-1) = \frac{2\pi}{r}$$

At radius r = 0.38m, the current is –

$$I = \frac{2\pi}{0.38} = 16.53\,A$$

Hence, the correct option is (A)

Previous Year's Questions

Evaluate the flux density at the boundary when the charge density is given by 25 units.

(A) 12 (B) 25

(C) 48 (D) 96

RELAXATION TIME

- Homogeneous medium is a medium where the physical features do not vary in the medium. They remain same from point to point in the medium throughout whereas, if the characteristics of medium vary from one point to another, then the medium is regarded as a non-homogeneous medium or heterogeneous medium.
- The medium is called linear with respect to the electric field, if the flux density ($\vec{D}$) is directly proportional to the electric field ($\vec{E}$).

  While, non-linear medium/material is where the flux density is not directly proportional to $\vec{E}$ in the material.
- Assume a conducting material, which is linear and has a homogeneous medium.

  Therefore, the current density ($\vec{J}$) for such material will be-

  $$\vec{J} = \sigma\vec{E}$$

  and $\vec{D} = \in \vec{E}$ (for linear material)

Gray Matter Alert!!!

Relaxation time (τ) is a time interval between two consecutive collisions of electrons in a conductor when current flows through it.

$$\therefore \vec{E} = \frac{\vec{D}}{\epsilon}$$

Thus, $\vec{J} = \sigma \frac{\vec{D}}{\epsilon} = \frac{\sigma}{\epsilon}\vec{D}$ ... (3.12)

- The point form of continuity equation states that-

$$\nabla \cdot \vec{J} = -\frac{\partial \rho_v}{\partial t}$$

$$\therefore \quad \nabla \cdot \left(\frac{\sigma}{\epsilon}\vec{D}\right) = -\frac{\partial \rho_v}{\partial t}$$

$$\frac{\sigma}{\epsilon}(\nabla \cdot \vec{D}) = -\frac{\partial \rho_v}{\partial t}$$

But we know, $\nabla \cdot \vec{D} = \rho_v$

So, $\frac{\sigma \rho_v}{\epsilon} = -\frac{\partial \rho_v}{\partial t}$

$$\therefore \quad \frac{\partial \rho_v}{\partial t} + \left(\frac{\sigma}{\epsilon}\right)\rho_v = 0 \qquad \text{...(3.13a)}$$

- The above equation is homogeneous linear differential equation and by variable separation method, we get -

$$\frac{\partial \rho_v}{\rho_v} = -\left(\frac{\sigma}{\epsilon}\right)\partial t$$

On integrating both sides, we get -

$$\ln \rho_v = -\frac{\sigma t}{\epsilon} + \ln \rho_{v0}$$

where ln ρ_{v0} = constant of integration

$$\therefore \rho_v = \rho_{v0} e^{-\left(\frac{t}{\tau}\right)}$$

where $\tau = \frac{\epsilon}{\sigma}$ = time constant (seconds)

$$\rho_v = \rho_{v0} e^{-\left(\frac{\sigma}{\epsilon}\right)t} = \rho_{v0} e^{-\left(\frac{t}{\tau}\right)} \qquad \text{...(3.13b)}$$

where ρ_{v0}= initial charge density at (t = 0).

Rack your Brain



At the dielectric-dielectric boundary, which component of the electric-field intensity is always continuous?

Gray Matter Alert!!!

These boundary conditions implies the relationships of the vector field only at the dielectric interface. They do not imply anything about the value of fields at points above or below the interface.

This time τ is referred to as the relaxation time.

- Equation (3.13b) shows that if the charge is introduced at some interior point of the material then ρ_v will decay exponentially.
- Relaxation time (τ) is the time taken by a charge placed inside the material to drop to e^{-1} (e^{-1} = 36.8%) of the initial value.

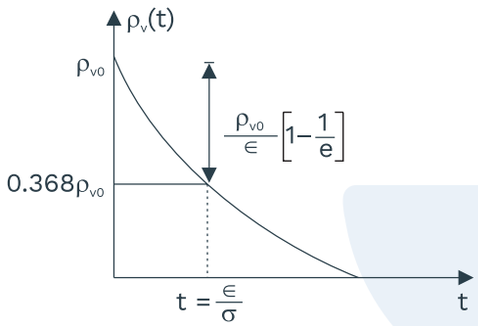


Figure 3.2

- Relaxation time is generally shorter for good conductors while for the good dielectrics, it is longer.

For copper (good conductor), $\sigma_{Cu} = 5.8 \times 10^7$ S/m

$\epsilon_r = 1$

$$\therefore \tau = \frac{\epsilon}{\sigma} = \frac{\epsilon_r \epsilon_0}{\sigma} = 1 \times \frac{10^{-9}}{36\pi} \times \frac{1}{5.8 \times 10^7}$$

$\tau = 1.53 \times 10^{-19}$ sec

For fused quarts (good dielectrics)

$\sigma = 10^{-17}$ S/m and $\epsilon_r = 5$

τ = 51.2 days

BOUNDARY CONDITION

- When an electric field goes from one medium to another, boundary conditions are the conditions that occur at the boundary of the two media.
- The boundary conditions are of two types depending upon the nature of the media.

(i) Boundary condition between a conductor and free space.

(ii) Boundary condition between the two dielectrics with different properties.

Rack your Brain

What are the boundary conditions for the magnetic fields?

Previous Year's Questions

The electric field intensity is given 25 units for a surface with permittivity $\epsilon_r = 2.5$. What will be the field intensity of the surface in the air?

(A) 5.14 (B) 0.194

(C) 62.5 (D) 29

(iii) Boundary condition between the conductor and dielectric interface.

- To analyze the boundary conditions, the Maxwell's equations of electrostatics is required. So,

$$\oint \vec{E} \cdot d\vec{l} = 0$$

and $$\oint \vec{D} \cdot d\vec{S} = Q_{enc}$$

where Q_{enc} = free charge enclosed by the surface.

- The electric field intensity ($\vec{E}$) is decomposed into two components i.e., normal component ($\vec{E}_n$) to the boundary and tangential component ($\vec{E}_t$).

The total electric field ($\vec{E}$), will be,

$$\vec{E} = \vec{E}_t + \vec{E}_n \quad \text{... (3.15)}$$

1. Dielectric- Dielectric boundary conditions

- Consider the boundary between two perfect dielectrics where dielectrics have permittivities $\in_1$ and $\in_2$.

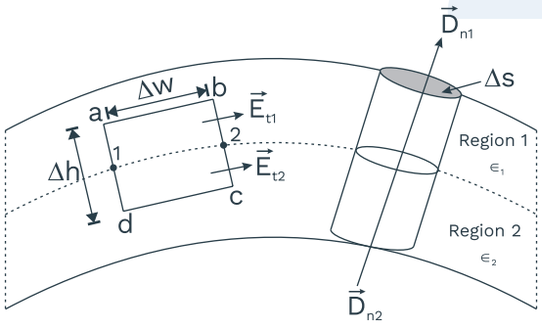


Figure 3.3

- The components of $\vec{E}$ and $\vec{D}$ are resolved into tangential and normal components to the surface.
- The tangential components of the electric field intensity ($\vec{E}$) at the boundary of both dielectrics remains the same i.e., electric field intensity is continuous across dielectric-dielectric boundary.

Rack your Brain

A conducting sphere with a charge of 'Q' is half-submerged in a non-conducting dielectric liquid with a dielectric constant of $\in$. The second half of the spherical is floating in mid-air. Will there be a purely radial electric field? (Yes/No)

- Mathematically,

$$\vec{E}_{t1} = \vec{E}_{t2} \qquad \ldots(3.16)$$

Also, $\vec{D}_{t1}$ and $\vec{D}_{t2}$ are magnitudes of tangential components of $\vec{D}$ in dielectric 1 and dielectric 2 respectively.

- Thus,

$$\vec{D}_{t1} = \epsilon_1 \vec{E}_{t1}$$

and $$\vec{D}_{t2} = \epsilon_2 \vec{E}_{t2}$$

Therefore, $$\frac{\vec{D}_{t1}}{\vec{D}_{t2}} = \frac{\epsilon_1}{\epsilon_2} = \frac{\epsilon_{r_1}}{\epsilon_{r_2}} \qquad \ldots(3.17)$$

- The tangential components of $\vec{D}$ varies across the interface/boundary.
- If no charge exists at the interface, the normal components of $\vec{D}$ is -

$$\vec{D}_{n1} = \vec{D}_{n2} \qquad (\rho_s = 0) \qquad \ldots(3.18)$$

The normal components of $\vec{D}$ is continuous at boundary between two perfect dielectrics.

Thus, $\vec{D}_{n1} = \epsilon_1 \vec{E}_{n1}$

$$\vec{D}_{n2} = \epsilon_2 \vec{E}_{n2}$$

Therefore,

$$\frac{\vec{D}_{n1}}{\vec{D}_{n2}} = \frac{\epsilon_1 \vec{E}_{n1}}{\epsilon_2 \vec{E}_{n2}} = 1 \quad (\text{Since, } \vec{D}_{n1} = \vec{D}_{n2}) \qquad \ldots(3.19)$$

$$\therefore \quad \frac{\vec{E}_{n1}}{\vec{E}_{n2}} = \frac{\epsilon_2}{\epsilon_1} = \frac{\epsilon_{r_2}}{\epsilon_{r_1}} \qquad \ldots(3.20)$$

- The above equation (3.20) shows that the normal components of electric field intensity ($\vec{E}$) are inversely proportional to the relative permittivity of two media.

Gray Matter Alert!!!

The boundary condition helps to determine the electric field above the material with surface charge. The electric field can be determined even if an external electric field is present.

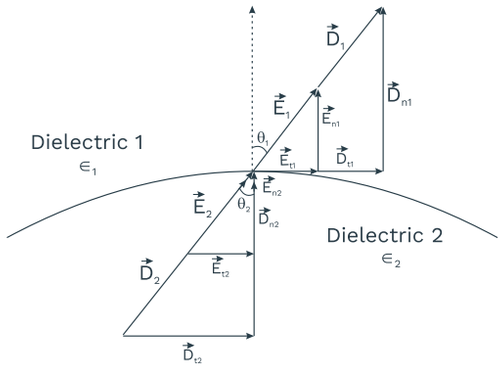


Figure 3.4

- Considering $\vec{D}_1$ or $\vec{E}_1$ and $\vec{D}_2$ or $\vec{E}_2$ making angles θ_1 and θ_2 with normal to the interface.

$$\vec{E}_{t1} = \vec{E}_1 \sin\theta_1$$

$$\vec{E}_{t2} = \vec{E}_2 \sin\theta_2$$

As $\vec{E}_{t1} = \vec{E}_{t2}$

$$\vec{E}_1 \sin\theta_1 = \vec{E}_2 \sin\theta_2 \qquad ...(3.21)$$

Similarly,

$$\vec{D}_{n1} = \epsilon_1 \vec{E}_1 \cos\theta_1$$

$$\vec{D}_{n2} = \epsilon_2 \vec{E}_2 \cos\theta_2$$

As $\vec{D}_{n1} = \vec{D}_{n2}$

$$\epsilon_1 \vec{E}_1 \cos\theta_1 = \epsilon_2 \vec{E}_2 \cos\theta_2 \qquad ...(3.22)$$

From equation (3.21) and (3.22) we have

$$\frac{\tan\theta_1}{\epsilon_1} = \frac{\tan\theta_2}{\epsilon_2} \qquad ...(3.23)$$

As we know, $\epsilon_1 = \epsilon_0 \epsilon_{r1}$
and $\epsilon_2 = \epsilon_0 \epsilon_{r2}$

- As $\rho_s = 0$, this is the law of refraction of the electric field at the boundary free of charge.
- In general, a dielectric interface causes flux lines to bend, resulting in uneven polarization charges accumulating on the opposite side.

Previous Year's Questions

Which one of the following is CORRECT for the Laplace equation: $\nabla^2 A=0$?

(A) Solutions are not continuous.
(B) Solutions are not separable in the coordinates.
(C) Solutions are not dependent on the boundary conditions.
(D) Solutions have maxima and minima at the boundaries.

Rack your Brain

What is difference between homogeneous and isotropic medium?

2. Conductor- Dielectric boundary conditions

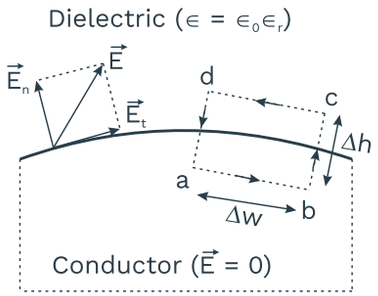


Figure 3.5 (a)

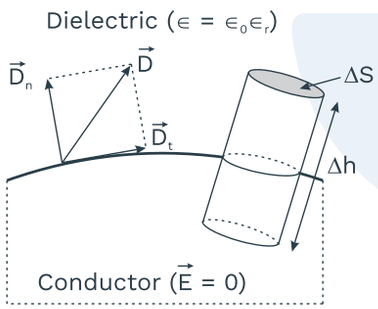


Figure 3.5 (b)

- Figure (3.5(a) and 3.5(b)) shows the cases of conductor–dielectric boundary equations.
- The conductor is considered to be perfect with $\sigma \to \infty$ and $\rho_c \to 0$.
- For determining the boundary condition, applying the fact $\vec{E} = 0$ inside the conductor.

$$0 \cdot \Delta w + 0 \cdot \frac{\Delta h}{2} + \vec{E}_n \cdot \frac{\Delta h}{2} - \vec{E}_t \cdot \Delta w - \vec{E}_n \cdot \frac{\Delta h}{2} - 0 \cdot \frac{\Delta h}{2} = 0$$

...(3.25)

As $\Delta h \to 0 \quad \therefore \quad \vec{E}_t = 0$

Similarly, by applying $\oint_S \vec{D} \cdot d\vec{S} = Q_{enc}$ to cylindrical part, $\Delta h \to 0$, we have -

$$\Delta Q = \vec{D}_n \cdot \Delta S - 0 \cdot \Delta S \qquad ...(3.26)$$

because $\vec{D} = \epsilon \vec{E} = 0$ inside the conductor.

Previous Year's Questions

A 2nF parallel-plate capacitor is charged to an initial potential difference (ΔV_i) = 200 V and then isolated. The dielectric material between the plates is mica, with a dielectric constant of 5.
How much work is required to withdraw the mica sheet?

(A) 4×10^{-5}J
(B) 0.4×10^{-5}J
(C) 8×10^{-5}J
(D) 2×10^{-5}J

Rack your Brain

The potential difference between two parallel plates 1.28 cm apart is 175V. The electric field strength between them is_____V/m.

$\therefore$ Above equation (3.26) can be re-written as

$$\vec{D}_n = \frac{\Delta Q}{\Delta S} = \rho_s \text{ or } \vec{D}_n = \rho_s \quad \ldots(3.27)$$

Example 2:
The two different dielectrics are separated by y = 0 plane. The surface charge density ρ_s = 4nC/m^2 and the dielectric in medium-1 (y > 0) has $\epsilon_1 = 2\epsilon_0$ and medium-2 is a perfect conductor. The electric field at point (3, 1, 4) will be -

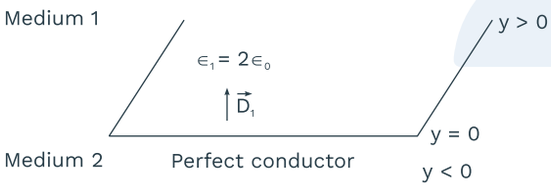
(A) $0.22 \times 10^{12} \hat{a}_x$

(B) $0.5 \times 10^{12} \hat{a}_y$

(C) $0.25 \times 10^{12} \hat{a}_x$

(D) $0.225 \times 10^{3} \hat{a}_y$

Solution:
Medium-1 is a dielectric medium with $\epsilon_1 = 2\epsilon_0$ and medium-2 is a perfect conductor.
The surface charge density, ρ_s = 4 nC/m^2



As the given point A(3, 1, 4) will be in medium-1 region.

$\therefore \vec{D}_1 = \rho_s \hat{a}_n$

Or $\vec{D}_1 = 4\text{nC/m}^2 \hat{a}_y$

The electric field will be –

$$\vec{E}_1 = \frac{\vec{D}_1}{\epsilon_1} = \frac{4 \times 10^{-9}}{2 \times 8.85 \times 10^{-12}} \hat{a}_y = 0.225 \times 10^{3} \hat{a}_y$$

Thus, the correct option is (D)

Previous Year's Questions

What will happen if a parallel plate capacitor is charged and disconnected from the battery after some time. Also, the plates were now moved away by pulling insulating handles,

(A) The energy stored in the capacitor will decrease.
(B) The capacitance of the capacitor will increase.
(C) The charges on the capacitor will decrease.
(D) The voltage across the capacitor will increase.

Rack your Brain

What is the significance of boundary conditions?

3. Conductor- Free space boundary condition

- The conductor-free space boundary conditions are a special case of conductor-dielectric conditions.
- The boundary conditions of this interface can be obtained by replacing $\epsilon_r = 1$ for free space (or special dielectric).
- Thus, the electric field $\vec{E}$ must be normal to its surface and external to the conductor.

Thus, the boundary conditions are-

$$\vec{D}_t = \epsilon_0 \vec{E}_t = 0$$

and $$\vec{D}_n = \epsilon_0 \vec{E}_n = \rho_s \quad ...(3.28)$$

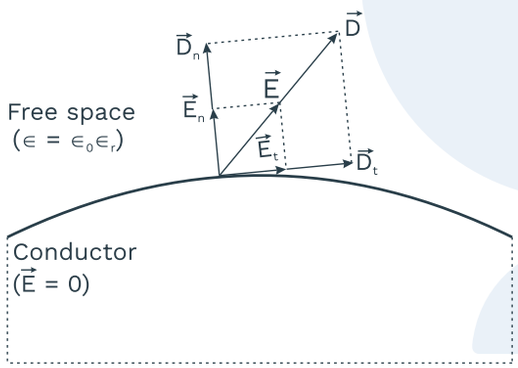


Figure 3.6

- $\vec{E}_n$ are inversely proportional to the relative permittivities of the two media. The magnitude of $\vec{D}_2$ can be obtained as

$$D_2 = D_1\sqrt{\cos^2\theta_1 + \left(\frac{\epsilon_2}{\epsilon_1}\right)^2 \sin^2\theta_1} \quad ...(3.29)$$

- Similarly, the magnitude of $\vec{E}_2$ can be obtained as

$$E_2 = E_1\sqrt{\sin^2\theta_1 + \left(\frac{\epsilon_1}{\epsilon_2}\right)^2 \cos^2\theta_1} \quad ...(3.30)$$

The above equations show that -

- $\vec{D}$ is larger in the region of larger permittivity.
- $\vec{E}$ is larger in the region of smaller permittivity.

Previous Year's Questions

Two capacitors have a distance between the plates equal to d, potential difference between the plates V, $\epsilon_1 = 1$, and $\epsilon_2 = 2$ and the area of each plate for the first capacitor is S and the electric field between the plates is $\vec{E}_1$. For the second capacitor, the area of each plate is twice the area of the first capacitor plates and the electric field between the plates $\vec{E}_2$. Analyze the relation between both electric fields, $\vec{E}_1$ and $\vec{E}_2$ -

(A) $\vec{E}_2 = \frac{\vec{E}_1}{4}$ (B) $\vec{E}_2 = \frac{\vec{E}_1}{2}$

(C) $\vec{E}_1 = \vec{E}_2$ (D) $\vec{E}_2 = 6\vec{E}_1$

$|\vec{D}_1| = |\vec{D}_2|$, if $\theta_1 = \theta_2 = 0°$

$|\vec{E}_1| = |\vec{E}_2|$, if $\theta_1 = \theta_2 = 90°$

Example 3:
Consider the two extensive dielectrics that are separated by z = 0 plane. For z > 0, the relative permittivity $\epsilon_{r_1} = 6$ and for $z < 0$, $\epsilon_{r_2} = 3$. The electric field in the medium-1 is a uniform field and is given as $\vec{E} = 5\hat{a}_x - 6\hat{a}_y + 8\hat{a}_z$ kV/m. The electric field for medium-2 will be -

(A) $6\hat{a}_x + 8\hat{a}_y - 5\hat{a}_z$

(B) $5\hat{a}_x + 6\hat{a}_y - 4\hat{a}_z$

(C) $5\hat{a}_x - 6\hat{a}_y + 16\hat{a}_z$

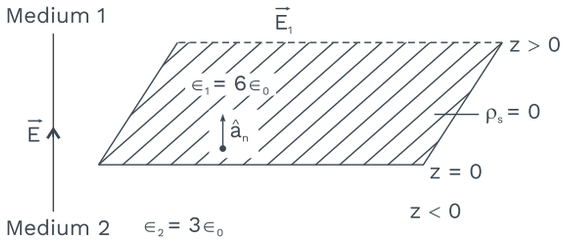
(D) $6\hat{a}_x - 5\hat{a}_y + 8\hat{a}_z$

Solution:
Given,
For medium-1, the electric field is –

$\vec{E} = 5\hat{a}_x - 6\hat{a}_y + 8\hat{a}_z$ kV/m and $\epsilon_{r_1} = 6$ $(\text{for } z > 0)$

and $\epsilon_{r_2} = 3$ $(\text{for } z < 0)$.

According to the condition given –



$$\hat{a}_n = \frac{\nabla V}{|\nabla V|} = \hat{a}_z \qquad \{\because z = 0\}$$

Also, the electric field is moving from medium-2 to medium-1.

and $\vec{E}_1 = 5\hat{a}_x - 6\hat{a}_y + 8\hat{a}_z$

Gray Matter Alert!!!

Charges create electric field lines pointing away from the positive charges and towards the negative charges. In a parallel-plate capacitor, the electric field lines point straight to the gap in between the two plates.

Rack your Brain

Consider a charged and isolated parallel plate capacitor. What is the effect of increasing the plate separation on a potential, capacitance, respectively?
(i) Decreases, decreases
(ii) Decreases, decreases
(iii) Decreases, increases
(iv) Increases, decreases

After applying boundary conditions,

$$\vec{E}_{n1} = \left(\vec{E}_1 \cdot \hat{a}_n\right) \cdot \hat{a}_n = 8\hat{a}_z$$

and $\vec{E}_{t2} = \vec{E}_{t1} = 5\hat{a}_x - 6\hat{a}_y$

Also $\vec{D}_{n1} = \vec{D}_{n2}$

$$\therefore \epsilon_1 \vec{E}_{n1} = \epsilon_2 \vec{E}_{n2}$$

$$\vec{E}_{n2} = \frac{\epsilon_1}{\epsilon_2}\vec{E}_{n1} = \frac{6}{3}\left(8\hat{a}_z\right)$$

$$\vec{E}_{n2} = 16\hat{a}_z$$

$$\therefore \vec{E}_2 = \vec{E}_{t2} + \vec{E}_{n2}$$

$$\vec{E}_2 = 5\hat{a}_x - 6\hat{a}_y + 16\hat{a}_z \text{ kV/m}$$

Thus, the correct option is (C).

Example 4:

The angles made by the electric fields $\vec{E}_1$ and $\vec{E}_2$ with the surface z = 0 if $\vec{E}_1 = 5\hat{a}_x - 6\hat{a}_y + 8\hat{a}_z$ and $\vec{E}_2 = 5\hat{a}_x - 6\hat{a}_y + 16\hat{a}_z$ are-

(A) $\alpha_1 = 26°$; $\alpha_2 = 60°$
(B) $\alpha_1 = 45°$; $\alpha_2 = 60°$
(C) $\alpha_1 = 45.69°$; $\alpha_2 = 63.99°$
(D) $\alpha_1 = 30°$; $\alpha_2 = 45°$

Solution:

We know that,

$$\vec{E}_1 = 5\hat{a}_x - 6\hat{a}_y + 8\hat{a}_z$$

Here $\vec{E}_{t1} = 5\hat{a}_x - 6\hat{a}_y$

$$\vec{E}_{n1} = 8\hat{a}_z$$

$$\therefore \tan\theta_1 = \frac{\left|\vec{E}_{t1}\right|}{\left|\vec{E}_{n1}\right|} = \frac{\sqrt{61}}{8}$$

$$\theta_1 = \tan^{-1}\left[\frac{\sqrt{61}}{8}\right]$$

$$\theta_1 = 44.31°$$

Rack your Brain

If the potential difference between the conductors is doubled. Then, the magnitude of the charge on each plate of the capacitor will be-
(i) four times larger.
(ii) two times larger.
(iii) unchanged.
(iv) one-half as large.

The angle from the surface is-

$\alpha_1 = 90° - \theta_1 = 90° - 44.31$

$\alpha_1 = 45.69°$

where, θ = angle between electric field and normal.

α = angle between electric field and surface.

For medium-2 with the electric field $\vec{E}_2$,

$$\vec{E}_2 = 5\hat{a}_x - 6\hat{a}_y + 16\hat{a}_z$$

Here $\vec{E}_{t1} = 5\hat{a}_x - 6\hat{a}_y$ and $\vec{E}_{n1} = 16\hat{a}_z$

$$\therefore \tan\theta_2 = \frac{|\vec{E}_{t1}|}{|\vec{E}_{n1}|} = \frac{\sqrt{61}}{16}$$

$$\therefore \theta_2 = \tan^{-1}\left[\frac{\sqrt{61}}{16}\right]$$

$$\theta_2 = 26.01°$$

Here, the angle between $\vec{E}_2$ and the surface will be α_2.

$\alpha_2 = 90° - \theta_2$

$\alpha_2 = 90° - 26.01°$

$\alpha_2 = 63.99°$

Hence, the correct option is (C).

POISSON'S AND LAPLACE'S EQUATION

- The Poisson's and Laplace's equations are used to analyze the charge and potential at some of the region's boundaries.
- If the charge distribution or potential throughout the region is given, using Poisson's and Laplace's equations will be easier.
- Poisson's equation is usually derived from the point form of Gauss's law. (For a linear, isotropic material medium).

$$\nabla \cdot \vec{D} = \nabla \cdot (\epsilon \vec{E}) = \rho_v \quad ...(3.31)$$

and $$\vec{E} = -\nabla V \quad ...(3.32)$$

Now, on substituting equation (3.32) into equation (3.31) we get -

Previous Year's Questions

The Poisson's equation for a homogeneous and an isotropic medium with a charge density ρ & dielectric constant ϵ, the electrical potential V will be:

(A) $\nabla^2 V = \frac{\rho}{\epsilon}$ (B) $\nabla^2 V = -\frac{\rho}{\epsilon}$

(C) $\nabla V = \frac{\rho}{\epsilon}$ (D) $\nabla V = -\frac{\rho}{\epsilon}$

Gray Matter Alert!!!

Poisson's equation allows no stability and Laplace's equation has no local maxima or minima of the potential.

Rack your Brain

Which of equation is a linear equation between Poisson equation and Laplace equation?

$$\nabla \cdot (-\in \nabla V) = \rho_v \quad \text{... (3.33)}$$

- For homogeneous medium, equation (3.33) becomes-

$$\nabla^2 V = -\frac{\rho_v}{\in} \quad \text{... (3.34)}$$

Equation (3.34) is referred as Poisson's equation.

- When $\rho_v = 0$ (i.e. for no charge region), equation (3.34) becomes Laplace equation as expressed below-

$$\nabla^2 V = 0 \quad \text{... (3.35)}$$

Recalling the Laplacian operator ∇^2 which was derived in chapter 1. Equation (3.35) reminds the expression of the Laplacian operator.

CAPACITOR

- Two (or more) conductors carrying equal but opposite charges are required in a capacitor.
- This means that all flux lines exiting one conductor must terminate on the other conductor's surface.
- Free space or a dielectric separates these plates.

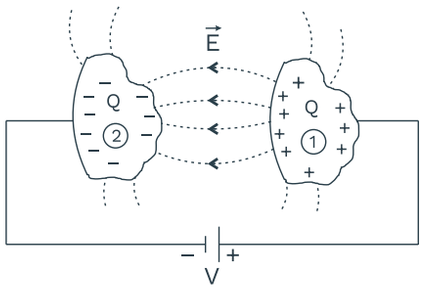


Figure 3.7

- Considering the two-conductor capacitor as shown in figure (3.7).
- The potential difference (V) maintained across the conductors is given by-

Gray Matter Alert!!!

Capacitors are very fast at charging and discharging (much faster than batteries). They are used for applications where rapid bursts of current are needed, such as camera flashes.

Definitions

Capacitors are energy-storing devices that consist of two plates of conducting material (generally a thin metal) which are separated by an insulator made of ceramic, film, glass, or other materials such as air. The insulator is also known as a dielectric, and it builds a capacitor's charging capacity.

$$V = V_1 - V_2 = -\int_2^1 \vec{E} \cdot d\vec{l} \qquad \ldots (3.36)$$

where $\vec{E}$ is the existing electric field between the conductors.

- Conductor-1 is assumed to carry positive charges while conductor-2 carries negative charges.
- The capacitor's capacitance C is equal to the ratio of the magnitude of the charge on one of the plates to the potential difference between the plates.
  Mathematically it can be expressed as-

$$C = \frac{Q}{V} = \frac{\epsilon \oint_S \vec{E} \cdot d\vec{S}}{\int_L \vec{E} \cdot d\vec{l}} \qquad \ldots (3.37)$$

Parallel-Plate capacitor

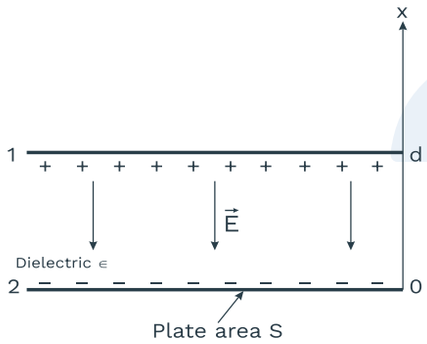


Figure 3.8

- Consider a parallel-plate capacitor as shown in the figure 3.8.
- Let us assume that each of the plates has an area S and separated by distance d.
- Suppose that plate-1 and plate-2 carry charges +Q and −Q uniformly distributed on them so that-

$$\rho_s = \frac{Q}{S} \qquad \ldots (3.38)$$

- An ideal parallel-plate capacitor has a very small separation distance (d) in comparison to the dimension of the plates.

Previous Year's Questions

Two insulating planar dielectric slabs having ϵ_1 and ϵ_2 are bonded together. Slab 1 has $\vec{E}_1$ at an angle θ_1 to the surface normal. What will be θ_2, if we assume the surface charge at the boundary $\rho_s = 0$?

(A) $\tan^{-1}[(\epsilon_2/\epsilon_1)\tan\theta_1]$
(B) $\tan^{-1}[(\epsilon_2/\epsilon_1)]$
(C) 0
(D) None of the above

- Considering such an ideal case, the fringing field at the edge of the plate as shown in the figure 3.9 below, is ignored because the field between them is uniform.

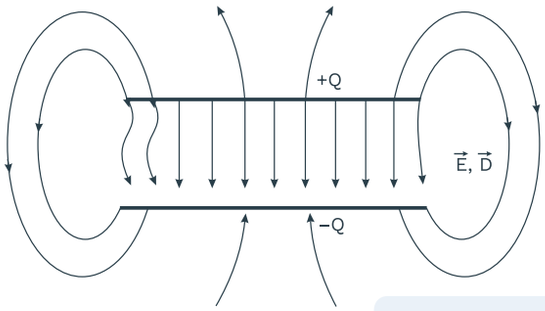


Figure 3.9: Fringing effect due to a parallel-plate capacitor

- If the space between the plates is filled with homogeneous dielectric with permittivity $\in$ and the fringing field at the edges of plates is ignored, then

$$\vec{D} = -\rho_s \hat{a}_x$$

or $$\vec{E} = \frac{-\rho_s}{\in} \hat{a}_x \quad ...(3.39)$$

Hence, $$V = -\int_2^1 \vec{E} \cdot d\vec{l} = -\int_0^d \left[-\frac{Q}{\in S} \hat{a}_x \right] \cdot dx \hat{a}_x \quad ..(3.40)$$

$$C = \frac{Q}{V} = \frac{\in S}{d} \quad ...(3.41)$$

- Equation (3.41) is the prime formula for measuring the capacitance C of a parallel-plate capacitor with dielectric filled in between the plates.
- The capacitance C_0 with air-filled between the plates of the capacitor is expressed as-

$$\in_r = \frac{C}{C_0} \quad ...(3.42)$$

- The energy stored in a capacitor is given by-

$$W_E = \frac{1}{2} CV^2 = \frac{1}{2} QV = \frac{Q^2}{2C} \quad ...(3.43)$$

Previous Year's Questions

Evaluate the conductivity (σ) when the current density is 10 units, and the electric field is 25 units. Also, identify the nature of the material.

(A) σ = 1.6 and Dielectric
(B) σ = 1.6 and Conductor
(C) σ = 0.4 and Dielectric
(D) σ = 0.4 and Conductor

Example 5:

Determine the ratio of energy density for the two extensive dielectrics separated by the x-y plane. If the given electric fields for $z > 0$ region and $z < 0$ regions are $7\hat{a}_x + 2\hat{a}_y + 3\hat{a}_z$ and $9\hat{a}_x - 8\hat{a}_y + 9\hat{a}_z$. And the relative permittivities for both mediums are 4 and 5, respectively.

(A) 0.21 (B) 0.1

(C) 0.3 (D) 1.1

Solution:

The electric field for medium-1 with $\in_{r_1} = 4$ is

$\vec{E}_1 = 7\hat{a}_x + 2\hat{a}_y + 3\hat{a}_z$

And the electric field for medium-2 with $\in_{r_2} = 5$ is

$\vec{E}_2 = 9\hat{a}_x - 8\hat{a}_y + 9\hat{a}_z$

We know that the energy densities for the mediums are given by –

$$W_{E_1} = \frac{1}{2} \in_1 \left|\vec{E}_1\right|^2 = \frac{1}{2} \in_0 \in_{r_1} \left|\vec{E}_1\right|^2$$

$$= \frac{1}{2} \times 4 \times 8.85 \times 10^{-12} \times 62$$

$$= 1097.4 \text{ pJ/m}^3$$

and $$W_{E_2} = \frac{1}{2} \in_2 \left|\vec{E}_2\right|^2$$

$$= \frac{1}{2} \times \in_0 \in_{r_2} \times \left|\vec{E}_2\right|^2$$

$$= \frac{1}{2} \times 8.85 \times 10^{-12} \times 5 \times 226$$

$$W_{E_2} = 5000.25 \text{ pJ/m}^3$$

The ratio of energy densities of medium-1 to medium-2 is-

$$\frac{W_{E_1}}{W_{E_2}} = 0.219$$

Hence, the correct option is (A).

Previous Year's Questions

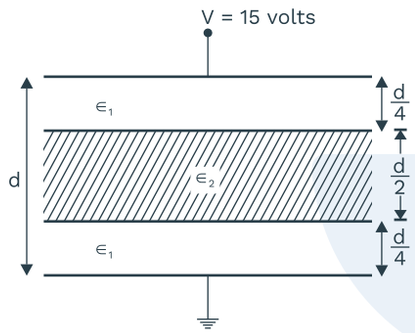
Two parallel metal plates are 2mm apart and have a uniform electric field strength 400 V m^{-1} between them. The potential difference between the plates is:

(A) 10.5 V (B) 16.7 V

(C) 8.1 V (D) 0.8 V

Example 6:

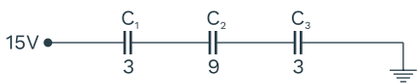
Consider a parallel plate capacitor having two dielectric materials. If the middle slab is placed in such a way that it is symmetrical with respect to the other two sides plates. The potential difference 3V is present between one of the middle plate and the nearest plate. The ratio of the permittivities will be –



(A) 3 : 2 (B) 2 : 3
(C) 9 : 4 (D) 4 : 9

Solution:

The equivalent circuit of the given figure will be –



$$C_1V_1 = C_2V_2$$

$$\frac{\in_0 \in_{r_1} A \times 3}{\frac{d}{4}} = \frac{\in_0 \in_{r_2} A \times 9}{\frac{d}{2}}$$

$$12 \in_{r_1} = 18 \in_{r_2}$$

$$\frac{\in_{r_1}}{\in_{r_2}} = \frac{18}{12}$$

$$\frac{\in_{r_1}}{\in_{r_2}} = \frac{3}{2}$$

Hence, the correct option is (A)

Multiple dielectrics between capacitor plates:

- **Case I:** When dielectrics have unequal width and plates have equal areas.

Rack your Brain

Why spherical capacitors are not widely used?

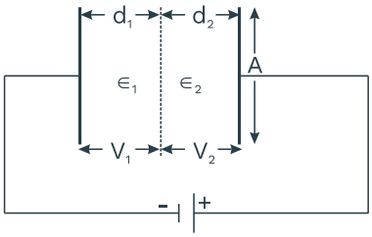


Figure 3.10

These two capacitors are in series as voltage divides.

$$C_{eq} = \frac{C_1 C_2}{C_1 + C_2}$$

- **Case II:** Unequal area of the plates and equal width of the dielectrics.

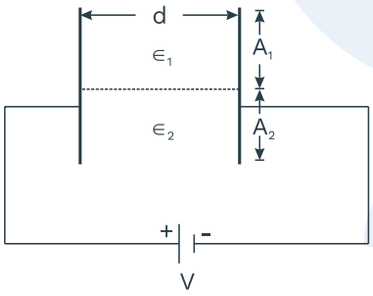


Figure 3.11

These two capacitors are in parallel and is expressed as

$$C_{eq} = C_1 + C_2$$

Coaxial capacitor

- Consider the length of two coaxial conductors as L, to be filled with homogeneous dielectric with ∈ permittivity.
- Assume that conductor 1 and conductor 2 carry +Q and –Q charges respectively distributed uniformly on the conductor surface.
  Applying Gauss's law to the arbitrary Gaussian cylindrical surface of radius ρ where (a < ρ < b), we get -

Rack your Brain

Under what conditions gases are a good conductor of electricity?

$$Q = \epsilon \oint_s \vec{E} \cdot d\vec{S} = \epsilon E_\rho 2\pi\rho L \quad \ldots(3.44)$$

Therefore, $\vec{E} = \frac{Q}{2\pi \epsilon \rho L}\hat{a}_\rho \quad \ldots(3.45)$

Ignoring the flux fringing at the cylinder ends-

$$V = -\int_2^1 \vec{E} \cdot d\vec{l} = -\int_b^a \left[\frac{Q}{2\pi \epsilon \rho L}\hat{a}_\rho\right] \cdot d\rho \cdot \hat{a}_\rho$$

$$V = \frac{Q}{2\pi \epsilon L}\ln\left(\frac{b}{a}\right) \quad \ldots(3.46)$$

Thus, the capacitance C of a coaxial cylinder is given by-

$$C = \frac{Q}{V} = \frac{2\pi \epsilon L}{\ln\left(\frac{b}{a}\right)} \quad \ldots(3.47)$$

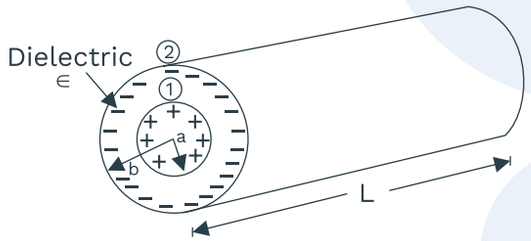


Figure 3.12: Coaxial capacitor

Rack your Brain

What is the significance of Wiedemann-Franz law?

Spherical capacitor

- A spherical capacitor is the case of two concentric spherical conductors.

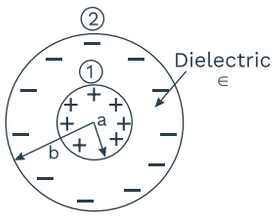


Figure 3.13: A spherical capacitor

- Consider the inner sphere of radius 'a' and outer sphere of radius 'b' where b > a.
  The two spheres are separated by dielectric medium with permittivity ∈.

Suppose +Q and –Q charges are distributed on the inner and outer spheres respectively. By applying Gauss's law to an arbitrary Gaussian spherical surface of radius r where a < r < b, then-

$$Q = \in \oint_s \vec{E} \cdot d\vec{S} = \in E4\pi r^2 \quad ...(3.48a)$$

$$\vec{E} = \frac{Q}{4\pi r^2 \in} \hat{a}_r \quad ...(3.48b)$$

- The potential difference between the conductor is-

$$V = -\int_2^1 \vec{E} \cdot d\vec{l} = -\int_b^a \left[\frac{Q}{4\pi \in r^2} \hat{a}_r\right] \cdot dr\, \hat{a}_r$$

$$V = \frac{Q}{4\pi \in}\left[\frac{1}{a} - \frac{1}{b}\right] \quad ...(3.49)$$

Therefore, the capacitance of the spherical capacitor is given by-

$$C = \frac{Q}{V} = \frac{4\pi \in}{\left(\frac{1}{a} - \frac{1}{b}\right)} \quad ...(3.50)$$

Recalling the expressions of resistance R and capacitance C of an electrical system.

$$R = \frac{V}{I} = \frac{\int_L \vec{E} \cdot d\vec{l}}{\int_S \sigma \vec{E} \cdot d\vec{S}} \quad ...(3.51)$$

$$C = \frac{Q}{V} = \frac{\in \oint_S \vec{E} \cdot d\vec{S}}{\int_L \vec{E} \cdot d\vec{l}} \quad ...(3.52)$$

- The product of equations (3.51) and (3.52) is expressed as-

$$RC = \frac{\in}{\sigma} \quad ...(3.53)$$

- For parallel-plate capacitor-

$$C = \frac{\in S}{d}, \quad R = \frac{d}{\sigma S}$$

Rack your Brain

What is the role of conductivity in the free-electron theory?

Previous Year's Questions

If two semi-infinite dielectric regions are separated by a plane boundary at y = 0.
The dielectric constants of region 1(y < 0) and region 2 (y > 0) are 2 and 5, respectively. Region 1 has uniform electric field $\vec{E} = 3\hat{a}_x + 4\hat{a}_y + 2\hat{a}_z$, where $\hat{a}_x, \hat{a}_y$ and $\hat{a}_z$ are unit vectors along the x, y and z axes, respectively. The electric field in region 2 is-

(A) $3\hat{a}_x + 1.6\hat{a}_y + 2\hat{a}_z$

(B) $1.2\hat{a}_x + 4\hat{a}_y + 2\hat{a}_z$

(C) $1.2\hat{a}_x + 4\hat{a}_y + 0.8\hat{a}_z$

(D) $3\hat{a}_x + 10\hat{a}_y + 0.8\hat{a}_z$

- For a cylindrical capacitor-

$$C = \frac{2\pi \in L}{\ln\left(\frac{b}{a}\right)} \text{ and } R = \frac{\ln\left(\frac{b}{a}\right)}{2\pi\sigma L}$$

- For a spherical capacitor-

$$C = \frac{4\pi \in}{\left(\frac{1}{a} - \frac{1}{b}\right)} \text{ and } R = \frac{\left(\frac{1}{a} - \frac{1}{b}\right)}{4\pi\sigma}$$

- For an isolated spherical conductor-

$$C = 4\pi \in a \text{ and } R = \frac{1}{4\pi\sigma a}$$

- Here, 'R' is the leakage resistance between the plates, not the resistance of capacitor plates, and σ is the conductivity of dielectric medium separating the plates.

Example 7:

Consider a capacitor made up of polymeric dielectric having $\in_r = 2.25$ and dielectric breakdown strength of 80 kV/cm. The rectangular plate of the capacitor has 40 cm length and 20 cm width. The maximum electric charge in the given capacitor will be –

(A) 13 mC (B) 12.74 C
(C) 10.13 μC (D) 12.74 μC

Solution:

Given,
Dielectric strength = 80 kV/cm.
Relative permittivity of polymeric dielectric

$\in_r = 2.25$

Rectangular plate capacitor has-
Length, l = 40 cm
Width, b = 20 cm
And we know that-
Q = CV = C(Ed)
Dielectric breakdown strength is the maximum electric field that a dielectric can tolerate above which dielectric goes into the breakdown.

Rack your Brain

What is Drude's theory of conductors?

Previous Year's Questions

A spherical capacitor has an inner sphere of radius 12 cm and an outer sphere of radius 13 cm. And the outer sphere is earthed, whereas the inner sphere has a charge of 2.5 μC. The space between the concentric spheres is filled with a liquid of dielectric constant 32. What will be the potential of the inner sphere?

(A) 4.5×10^2 V
(B) 7.5×10^2 V
(C) 1.5×10^{-2} V
(D) 4.5×10^{-2} V

∴ The capacitance will be –

$$C = \frac{\epsilon_0 \epsilon_r A}{d}$$

$$C = \frac{\epsilon_0 (2.25)\left(40 \times 20 \times 10^{-4}\right)}{d}$$

The maximum charge will be –

Q = C(Ed)

$$Q = \frac{\epsilon_0 (2.25)\left(40 \times 20 \times 10^{-4}\right)}{d} \times \frac{80 \times 10^3}{10^{-2}} \times d$$

$Q = 12.74 \times 10^{-6}$ C

Q = 12.74 μC

Hence, the correct option is (D)

Example 8:

Consider a coaxial cylindrical capacitor having dielectric relative permittivity as $\epsilon_{r_1} = 4$. What is the required relative permittivity (ϵ_{r_2}) when a quarter part of the dielectric is replaced by another dielectric in such a way that it doubles the capacitance?

(A) 20 (B) 10

(C) 8 (D) 5

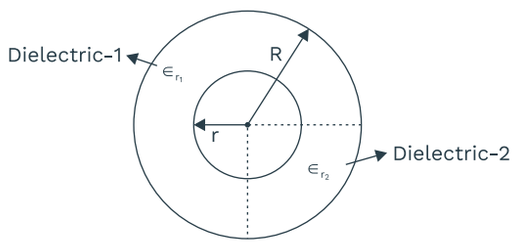
Solution:

Given,

The relative permittivity of the dielectric-1, $\epsilon_{r_1} = 4$.

Let the capacitance of the coaxial cylinder capacitor (before inserting dielectric-2) be C_0 and after inserting dielectric-2 in one-fourth part, the capacitance gets doubled.

i.e, $C_{eq} = 2C_0$



Gray Matter Alert!!!

Capacitance is the capability of a component or circuit to gather and store the energy in the form of an electrical charge.

Rack your Brain

What is Lorentz number?

$$\therefore C_0 = \frac{2\pi \epsilon_0 \epsilon_r L}{\ln\left(\frac{R}{r}\right)}$$

For shunt-connection,

$$C_{eq} = C_1 + C_2 = \frac{3C_{01}}{4} + \frac{C_{02}}{4}$$

$$C_{eq} = \frac{3}{4}\left[\frac{2\pi \epsilon_0 \epsilon_{r_1} L}{\ln\left(\frac{R}{r}\right)}\right] + \frac{1}{4}\left[\frac{2\pi \epsilon_0 \epsilon_{r_2} L}{\ln\left(\frac{R}{r}\right)}\right]$$

and $C_{eq} = 2C_0$ (According to the given condition)

$\therefore$ Substituting the values, we get –

$$2\left[\frac{2\pi \epsilon_0 \epsilon_{r_1} L}{\ln\left(\frac{R}{r}\right)}\right] = \frac{3}{4}\left[\frac{2\pi \epsilon_0 \epsilon_{r_1} L}{\ln\left(\frac{R}{r}\right)}\right] + \frac{1}{4}\left[\frac{2\pi \epsilon_0 \epsilon_{r_2} L}{\ln\left(\frac{R}{r}\right)}\right]$$

$$2\epsilon_{r_1} = \frac{3}{4}\epsilon_{r_1} + \frac{\epsilon_{r_2}}{4}$$

$$8\epsilon_{r_1} = 3\epsilon_{r_1} + \epsilon_{r_2}$$

$$\epsilon_{r_2} = 5\epsilon_{r_1} = 5 \times 4 = 20$$

$$\therefore \epsilon_{r_2} = 20$$

Thus, when the dielectric of $\epsilon_{r_2} = 20$ is placed at one-fourth portion of the co-axial cylindrical capacitor, then only its capacitance gets doubled. Hence, the correct option is (A).

Rack your Brain

What is the function of the second plate in a parallel-plate capacitor?

INDUCED CHARGES

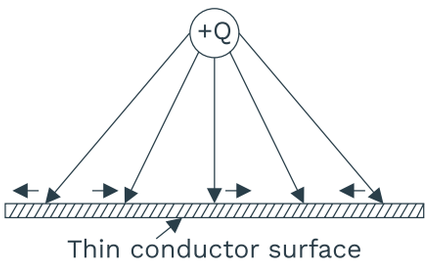


Figure 3.14

Gray Matter Alert!!!

The induced charge (-Q) can never be larger than the inducing charge.

- The field has tangential components and normal components. The tangential components of electron displaced on the conductor surface such that charges are periodically accumulated on the entire conductor surface.
- This is called an induced charge, such that the resultant field is a vector sum of $\vec{E}_1$ due to the actual charge and $\vec{E}_2$ due to the image charge, as shown in the figure 3.15.

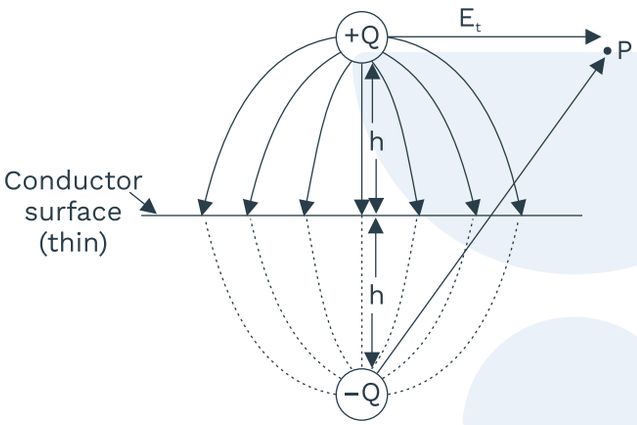


Figure 3.15

- The field has only normal components and is similar to the dipole field with a negative charge below the conductor surface. The negative charge is called an image charge and is the result of the induced charge.

METHOD OF IMAGES

- A dipole field and a plane is considered that exists midway between the two charges having zero potential infinite planes.
- The planes are represented by a very thin infinite conducting plane.
- The fields from the charge above the conducting plane can be obtained by replacing the conducting plane with an equipotential surface.

Definitions

In induction, a charged object is brought near without touching a neutral object. The presence of a charged object near a neutral conductor will induce (or force) the electrons within the conductor to move.

Rack your Brain

What are Dirichlet and Neumann boundary conditions in electrostatics?

Gray Matter Alert!!!

The method of images helps us to solve electrostatic boundary value problems without the Poisson and Laplace's equations, by replacing boundary surfaces with appropriate image charges.

- The charge at points above the conducting plane can be obtained by substituting an equipotential surface for the conducting plane. The original charge will be reversed in this image.
- If the charge is placed above a perfect conductor then the charge's image is formed beneath the equipotential surfaces, according to the method of image as shown below-

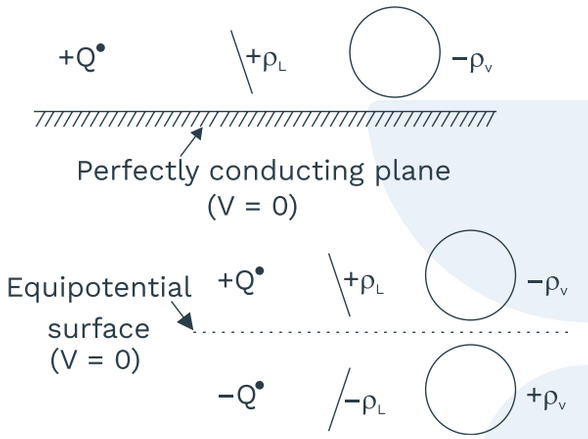


Figure 3.16

Rack your Brain

List some examples of charging by induction.

Definitions

In electrostatics, the image charges method is used to efficiently visualise the distribution of a charge's electric field in the area of a conducting surface.

- Condition for applying method of image-

(i) Image charges must be located in the conducting region.

(ii) The image charges must be located such that the potential must be zero or constant on the conducting surface.

- The first condition is to satisfy the Poisson's equation, while the other is to satisfy the boundary conditions.
- If a point charge is placed between two semi-infinite perfectly grounded conducting sheets placed at an angle θ degree, then the number of image charges is given by-

$$N = \left(\frac{360°}{\theta} - 1\right)$$

Example 9:

Which of the following situation involves the application of the method of images to boundary conditions in electrostatics ?

(A) When a set of conducting surfaces are added and the removal of the charge distributions are done.

(B) When a set of conducting surfaces are removed but some charge distributions are added.

(C) When both the charge distribution and set of conducting surfaces, are removed.

(D) When a set of conducting surfaces are removed but the addition of charge distribution is done.

Solution:

The image theory is generally employed in static fields only.

Also, when the charge distribution is replaced with the original charge distribution and their image charges.

The method of images can be used when the conducting surfaces are replaced by the equipotential surfaces.

Hence, the correct option is (D).

Example 10:

The method of images can be employed in which of the following fields?

(A) Only in electrostatic fields

(B) In electrodynamic fields only

(C) In electrostatic fields but not in electrodynamic fields.

(D) In both electrostatic fields and electrodynamic fields.

Gray Matter Alert!!!

Symmetry influences the use of image charge to satisfy boundary conditions on more than one planar surface. Even where symmetry is not involved, the charges that are located outside the region of interest to produce fields that satisfy boundary conditions are often referred to as image charges.

Rack your Brain

The charge within a conductor will be

(A) 1 (B) -1

(C) 0 (D) ∞

Solution:

The method of images (also known as method of image charge or method of mirror charge) is a basic problem solving tool in electrostatics.

The image theory defines that a given charge distribution lying above an infinite grounded perfect conducting plane can be replaced by the charge configuration itself.

Hence, the correct option is (A).

CHAPTER SUMMARY



- Materials are categorized as conductors ($\sigma >> 1$, $\epsilon_r = 1$) and dielectric ($\sigma << 1$, $\epsilon_r \geq 1$) in terms of the conductivity (σ)and the dielectric constant (ϵ) or relative permittivity.
- The flux of electric current density through a surface is defined as the electric current -

$$I = \int \vec{J} \cdot d\vec{S}$$

- For dielectric media, the $\vec{D}$ and $\vec{E}$ fields are related as $\vec{D} = \epsilon \vec{E}$, where $\epsilon = \epsilon_0 \epsilon_r$ is the permittivity of the medium while $\vec{E}$ and $\vec{P}$ are related as $\vec{P} = \chi_e \epsilon_0 \vec{E}$

  where,

  P = polarization density vector

  χ_e = electric susceptibility
- The electric susceptibility (χ_e) of dielectric measures the sensitivity of the material to an electric field.
- If $\vec{D} = \epsilon \vec{E}$ and if ϵ is independent of $\vec{E}$ then the dielectric material is linear.
- Medium is homogeneous, if ϵ is independent of the position.
- The law of charge conservation is the basis of Kirchhoff's current law, and is stated in the continuity equation-

$$\nabla \cdot \vec{J} + \frac{\partial \rho_v}{\partial t} = 0$$

- The relaxation time of the material is the time taken by a charge placed in its interior to decrease by the factor of e^{-1} or approximately 38% of its original magnitude.
- All boundary conditions must be achieved by an electric field existing in two different media which is separated by an interface.
- For a dielectric-dielectric interface, the boundary conditions are given by-

$$\vec{E}_{t1} = \vec{E}_{t2}$$

$$\vec{D}_{n1} - \vec{D}_{n2} = \rho_S \text{ or } \vec{D}_{n1} = \vec{D}_{n2}, \quad \text{if } \rho_S = 0$$

  And for a dielectric-conductor interface, the boundary conditions are given by-

$$\vec{E}_t = 0, \vec{D}_n = \epsilon \vec{E}_n = \rho_S$$

  because inside the conductor, the electric field is zero.

- An equipotential surface is a surface that exists with the same electrical potential at each point.
- The boundary value problems are generally solved by using Poisson's equation if $\rho_v \neq 0$ or by using Laplace's equation if $\rho_v = 0$.
- The Poisson's equation for a non-homogeneous region is given by-

$$\nabla \cdot \in (\nabla V) = - \rho_v$$

- The Poisson's equation for a homogeneous region, where $\in$ is independent of space variables is given by : $\nabla^2 V = -\frac{\rho_v}{\in}$.
- For a charge-free region ($\rho_v = 0$), the Poisson's equation becomes Laplace's equation and is expressed as $\nabla^2 V=0$.
- The capacitance (C) for the parallel-plate capacitor with dielectric filled in between the plates is given by -

$$C = \frac{Q}{V} = \frac{\in S}{d}$$

- The capacitance (C) of a coaxial cylinder is expressed as -

$$C = \frac{Q}{V} = \frac{2\pi \in L}{\ln\left(\frac{b}{a}\right)}$$

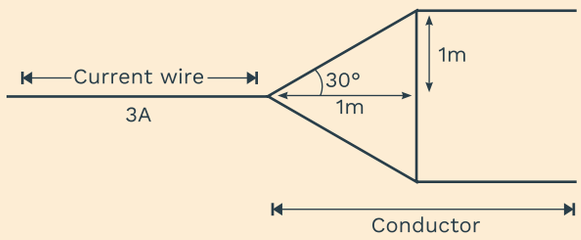
- The capacitance (C) of the spherical capacitor is expressed as-

$$C = \frac{Q}{V} = \frac{4\pi \in}{\left(\frac{1}{a} - \frac{1}{b}\right)}$$

- The image theory defines that the given charge configuration above an infinite grounded perfect conducting plane is substituted by the charge configuration itself or its image, and an equipotential surface in place of the conducting plane.
- For implementing the image method, the following conditions should be satisfied-
  The image charges must be situated inside the conducting region.
  The image charges must be situated in such a way that the potential is zero or constant on the conducting surfaces.
- The method of images (or method of mirror images) can be used in electrostatics (or magnetostatics), to visualize the distribution of the electric field (or magnetic field) of a charge (or magnet) in the area of the conducting surface.

PRACTICE QUESTIONS

Q1 **Consider a current carrying wire with current 3A placed with the conductor as shown in figure.**



The current density will be-

(A) 3 A/m **(B) 1.5 A/m** **(C) $\sqrt{3}$ A/m** **(D) $\frac{\sqrt{3}}{2}$ A/m**

Q2 **Consider the region x < 0 has electric field as $\vec{E}_1 = 5\hat{a}_x + \hat{a}_y$ and $\epsilon_{r_1} = 8$. Also, the x > 0 region with $\epsilon_{r_2} = 4$. If the fields in each region makes an angle of α_1 and α_2 with the normal to the boundary then-**

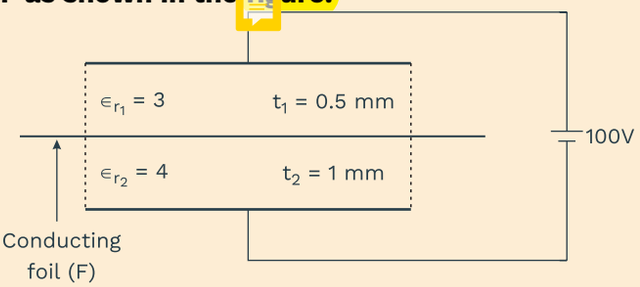
(A) $\alpha_1 = \alpha_2$

(B) $\alpha_1 < \alpha_2$

(C) $\alpha_1 > \alpha_2$

(D) Data insufficient

Q3 **A composite parallel plate capacitor is made up of two different dielectric materials with different thicknesses. The two dielectrics are separated by conducting foil F as shown in the figure.**



The voltage of the conducting foil will be -

(A) 52 V **(B) 68 V** **(C) 55 V** **(D) 60 V**

Q4 Consider the parallel plate capacitor with C_0 as capacitance having air as dielectric. If the half is filled with the dielectric($\in_r$), later will have modified capacitance will be-

(A) $\frac{1+\in_r}{C_0}$ (B) 0 (C) $\frac{C_0 \in_r}{2}$ (D) $C_0(1-\in_r)$

Q5 Consider a capacitor with the two metal plates each having 500 × 500 mm^2 area and separated by a distance of 6 mm. Assume that the space between the metal plates is filled with the glass plate (thickness = 4 mm, $\in_{glass}$ = 8) and a layer of paper (thickness = 2 mm, $\in_{paper}$ = 2). Also, neglect the fringing effect. The capacitance will be-

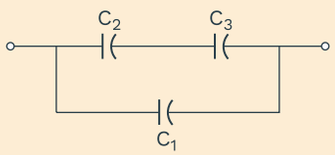
(A) 1400 pF (B) 1900 F (C) 700 pF (D) 1474 pF

Q6 The electric field in region-1 is given as 4 kV/cm of a parallel plate capacitor which is filled with the dielectrics $\in_{r_1} = 1$ and $\in_{r_2} = 4$. The distance between the plates of a capacitor is 2 cm then evaluate the electric field in region-2.



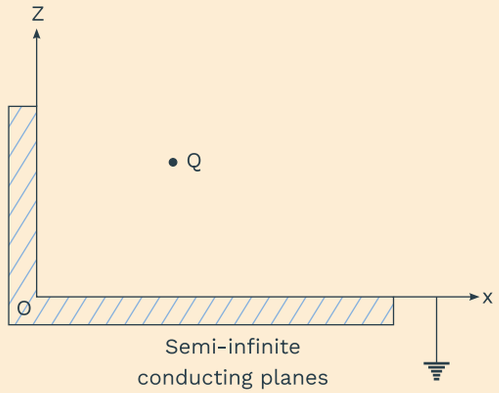
(A) 4 kV/cm (B) 2 kV/cm (C) 0.4 kV/m (D) 4 kV/m

Q7 Consider the 10 μF, 5 μF and 2 μF capacitors as C_1, C_2 and C_3 respectively. These capacitors has the breakdown voltages of 10V, 5V and 2V respectively. The maximum safer voltage that may be applied across the combination and corresponding total charge that can be stored in the effective capacitance across the terminals are-



(A) 4V and 32 μC (B) 3.2 V and 32 μC
(C) 2.8 V and 32 μC (D) 2.5 V and 2.5 μC

Q8 **Consider the point charge Q of 10 nC which is placed 10 cm away from the point O and along the line bisecting at θ = 60°. The magnitude of the force on Q due to the charge induced on the conducting walls will be-**



(A) 60.54 μN (B) 30.27 μN (C) 60.54 N (D) 6.54 μN

Q9 **Consider the capacitor coaxial metal cylinders with radii a = 1mm and b = 5 mm. Assume that the space between the cylinder is filled with a dielectric with $\in_r$ = 3 (1 + p) relative permittivity where (a < p < b). The capacitance per meter will be-**

(A) 0.41 nF (B) 0.86 nF (C) 0.32 nF (D) 0.5 nF

Q10 **Evaluate the magnitude of the force of attraction between the charges and the plane. If the two point charges of 3 nC and –4nC are placed at (0,0,1) meter and (0,0,2) meter, respectively while considering an infinite conducting plane at z = 0.**

(A) 5.25 nN (B) 2.15 nF (C) 10.50 nN (D) 3 nF

ANSWER KEY:

Q1 (B)	**Q2** (C)	**Q3** (D)	**Q4** (C)	**Q5** (D)	**Q6** (A)
Q7 (C)	**Q8** (A)	**Q9** (C)	**Q10** (A)		